UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 2 2023/2024

SECP2613 SYSTEM ANALYSIS AND DESIGN

SECTION 01

ASSIGNMENT 1: PROJECT PLANNING

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Question 1

a. Defining the objectives of which aspects of the system should be improved. Determine the resources in terms of operational feasibility, technical feasibility and the economic feasibility.

b. The people who use the system are all willing to adopt a new system as the new system may improve efficiency.

c. In terms of cost, it would be a huge amount as said by Abu. However, it can change the impression of the firm which is now stodgy. The change can bring benefits to the company. A detailed cost benefit analysis shall be done.

d. According to Abu, the technology to update the system is available.

e. Yes, a full-blown systems study should be done. The objective is to improve the firm's impression as the leader in wholesale food distributing thus improving revenue. In terms of resources, the technology is available to meet the needs, the cost would be a big amount of money but would bring benefit in the future and the people using the system are willing to adopt a new system.

Question 2

Student Registration

1.0 Research

1.1 Conduct Needs Assessment

(Duration : 7 days, Predecessor : None)

1.1.1 Assess current student registration processes

1.1.2 Identify pain points

1.1.3 Gather requirements from stakeholders

1.2 Market Research

(Duration : 10 days, Predecessor : None)

1.2.1 Research existing student registration systems

1.2.2 Analyze their features and costs

1.2.3 User feedback to inform decision-making

1.3 Financial Analysis

(Duration : 5 days, Predecessor : 1.1 and 1.2)

1.3.1 Estimate the budget required for developing and implementing the new student registration system

1.3.2 Considering software licenses, hardware costs, and personnel expenses

2.0 Planning

2.1 Define Project Scope

(Duration : 5 days, Predecessor : 1.1, 1.2 and 1.3)

2.1.1 Clearly outline the objectives, deliverables, and constraints of the student registration project

2.2 Form Project Team

(Duration : 3 days, Predecessor : 2.1)

2.2.1 Assemble a cross-functional team with expertise in software development, user experience design, and project management

2.3 Develop Project Plan

(Duration : 10 days, Predecessor : 2.2)

2.3.1 Create a detailed project plan outlining tasks, timelines, resource allocation, and dependencies for the development

3.0 Development

3.1 Develop System Requirements

(Duration : 5 days, Predecessor : 2.3)

3.1.1 Define specific requirements such as registration form fields, course selection, payment processing, and user authentication.

3.2 Design System Architecture

(Duration : 7 days, Predecessor : 3.1)

3.2.1 Design the overall architecture of the student registration system, including database structure, user interface layout, and integration with existing systems

3.3 Develop Prototype

(Duration : 14 days, Predecessor : 3.2)

3.3.1 To demonstrate key features and functionalities.

3.3.2 Undergo testing and refinement before final implementation.

4.0 Testing

4.1 Conduct User Acceptance Testing

(Duration : 7 days, Predecessor : 3.3)

4.1.1 Engage end-users, such as students, faculty, and administrative staff, to test the student registration module.

4.1.2 Gather feedback on usability, functionality, and performance to identify and address any issues or concerns.

5.0 Implementation Phase

5.1 Prepare for System Deployment

(Duration : 5 days, Predecessor : 4.1)

5.1.1 Prepare the necessary infrastructure and resources, including server setup, software installation, and data migration.

5.2 Train Staff

(Duration : 7 days, Predecessor : 5.1)

5.2.1 Provide training sessions for staff members responsible for managing the student registration system.

6.0 Evaluation

6.1 Evaluate System Performance

(Duration : 5 days, Predecessor : 5.2)

6.1.1 Assess the performance of the student registration system in real-world scenarios, monitoring factors such as system responsiveness, transaction processing speed, and user satisfaction.

Question 3

Some of the differences which managers face with Traditional and with agile are :-

1. Customer Involvement

In traditional projects, the sequential division of the 7 phases often assigns different tasks to various team members. For instance, the analyst identifies and analyzes problems, while programmers handle system development and implementation. This structure often results in limited customer interaction, primarily with the analyst team, who then relay customer needs to the programmers. Consequently, customers are less involved in the development process.

In contrast, agile methodology emphasizes iterative and incremental development, where requirements and implementations evolve over time. Development occurs in rapid cycles, with each release showing incremental progress. This approach enables customers to visualize development progress and provide feedback during the development phase. Also, the development team can handle the project requirements changing as they occur in the sprint or projects through time.

2. Customer Satisfaction

In traditional project management, the project follows a sequential model, meaning the progress of the project flows in one direction. Customer involvement is typically limited to the initial stages which is identifying the problems and stating the system requirements. This sequential nature makes it challenging to accommodate changes requested by customers later in the process. If the customer suddenly wants to make a change after the development phase has started, it could necessitate going back to phase 1 or result in the system not meeting their requirements.

Conversely, in agile project management, customer involvement is extensive at each development stage. This allows customers to keep record of the progress and also provide the managers with updates or requirements of changes if needed. By promoting continuous collaboration and flexibility, agile projects enable teams to adapt to changing needs more effectively, ultimately improving customer satisfaction.

3. Project Delivery Time

Traditional project management typically involves detailed plans and schedules at the outset with the goal of delivering the project on time. However, due to its sequential nature, it can be challenging to adjust the schedule once the project enters the execution phase to accommodate changing needs or circumstances.

Agile project management provides greater flexibility in managing project delivery times through short iterative development cycles. Each iteration involves the team developing and

delivering a working software product, allowing the project team to adjust plans in a timely manner based on customer feedback and changing needs. This flexibility helps ensure on-time project delivery and better manage delivery times throughout the development process.

4. Time zone and work location:

In traditional project management, team members are often located in the same place or within similar time zones. This makes it easier for them to talk face-to-face and collaborate to make the project go smoothly, but may limit the ability to work effectively with team members in different time zones and in different physical locations.

In contrast, agile project management embraces remote collaboration and distributed teams. As technology and communication tools advance, agile teams can work effectively even if they're in different time zones and in different physical locations. This flexibility enables organizations to tap into a global talent pool, promote diversity and leverage expertise from around the world. However, it can be tricky to coordinate when people are in different locations or have different schedules. The managers should have proactive planning and effective use of collaboration tools in order to get through these challenges.

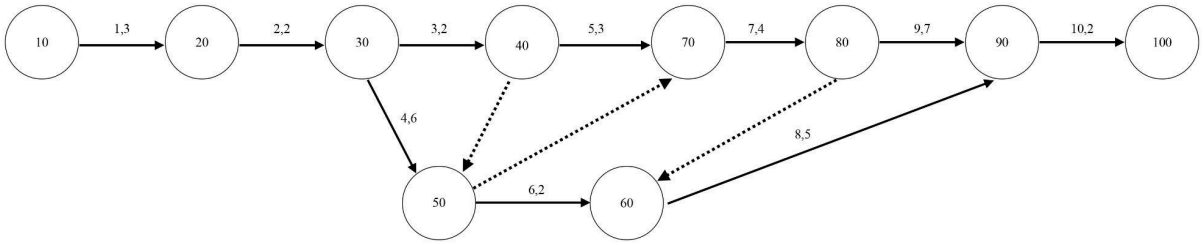
Question 4

Costs \ Year	0	1	2	3
Development Costs				
Hardware	50000			
Software	15000			
Training	30000			
Consulting	80000			
Data Conversion	40000			
Total	215000			
Production Costs				
Supplies		15000	15000	15000
IS Support		75000	82500	90750
Upgrades		10000	10000	10000
Annual Prod. Cost		100000	107500	115750
(Present Value)		74074	58985	47046
Accumulated Cost		289074	348059	395105
Benefits				
Improve Customer Service		150000	180000	216000
Increase Productivity		150000	187500	234375
Total		300000	367500	450375
(Present Value)		222222	201646	183051
Accumulated Benefits (PV)		222222	423868	606920
Gain or Loss		-66852	75809	211815
Profitability Index (PI)	0.99			

The profitability index is 0.99. We recommend continuing with the project as the PI value is almost 1.

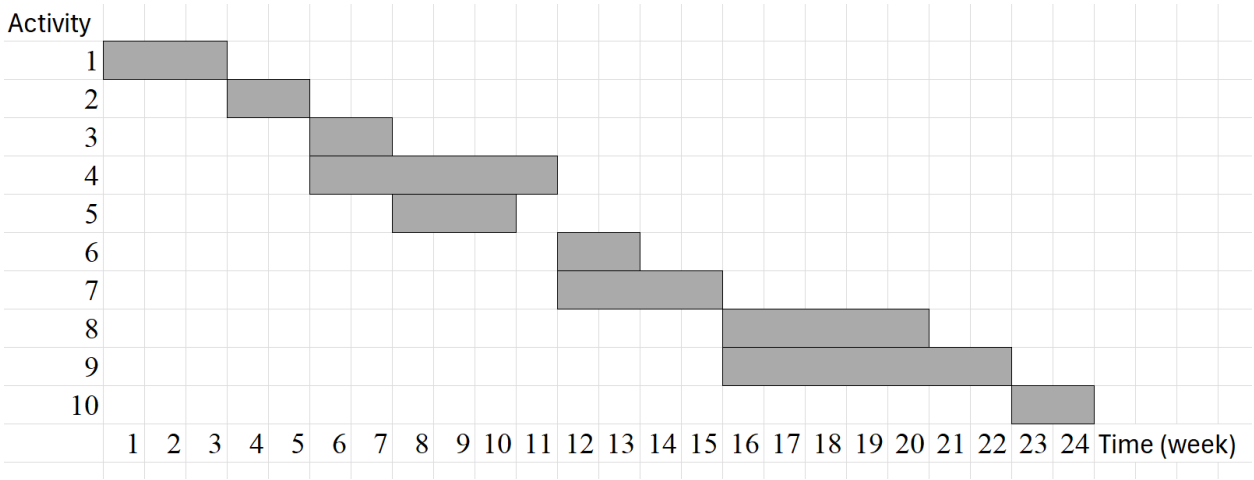
Question 5

a.



b. Path 1: 1-2-3-5-7-9-10
Path 2: 1-2-3-5-7-8-10
Path 3: 1-2-3-6-8-10
Path 4: 1-2-4-7-9-10
Path 5: 1-2-4-6-8-10
Path 6: 1-2-4-7-8-10

c. Path 1 length = 3+2+2+3+4+7+2 = 23
Path 2 length = 3+2+2+3+4+5+2 = 21
Path 3 length = 3+2+2+2+5+2 = 16
Path 4 length = 3+2+6+4+7+2 = 24
Path 5 length = 3+2+6+2+5+2 = 20
Path 6 length = 3+2+6+4+5+2 = 22

Since the critical path is the longest path through the network diagram, path 4, 1-2-4-7-9-10, is the critical path for this project.

d.